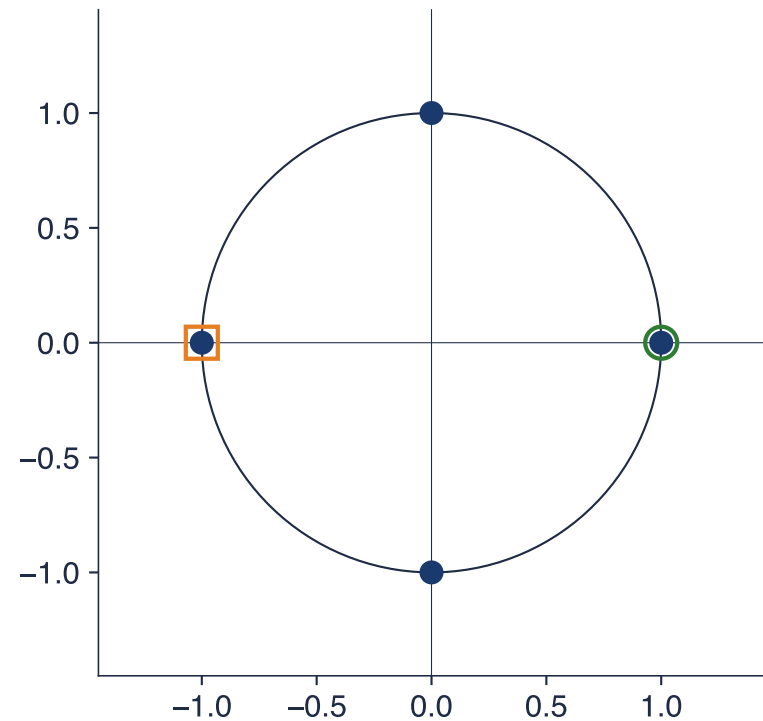
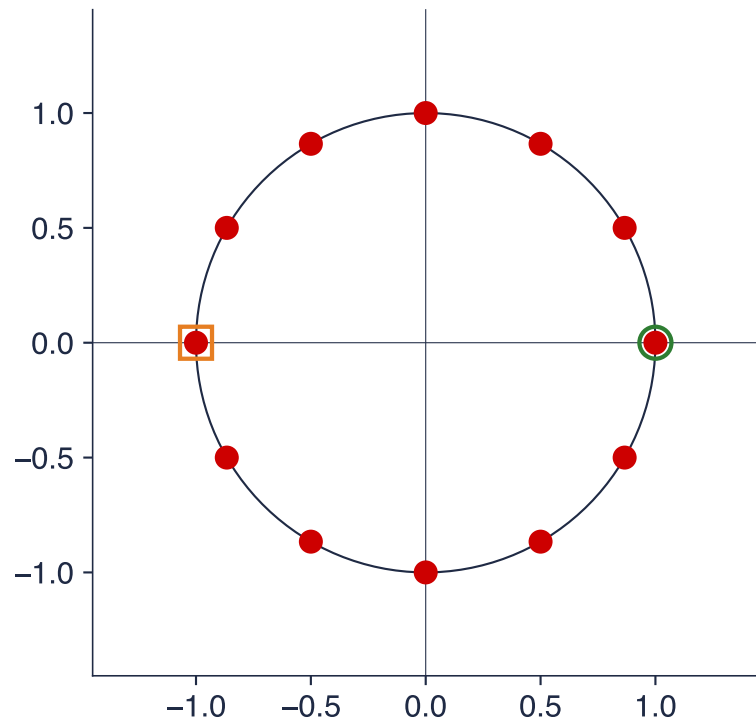


$s = 4$ : 4 roots of  $1 - z^4$



- roots of  $1 - z^4$
- $z = 1$ : zero frequency (trend)

$s = 12$ : 12 roots of  $1 - z^{12}$



- $z = -1$ : frequency 1/2 (period 2)
- roots of  $1 - z^{12}$